

User Manual

Version 1.0 · June 2026

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1 Introduction

EU Gas Storage Analysis is a Python + Jupyter toolkit for tracking European natural gas storage fundamentals, analysing TTF forward curve dynamics, and generating publication-ready market intelligence reports. It pulls live data from the GIE AGSI+ and ALSI+ APIs and the Databento TTF futures feed, then surfaces the analysis through 11 linked notebooks and a ReportLab PDF report.

Who it is for: energy traders monitoring EU injection/withdrawal pace and winter adequacy; analysts building storage-to-price regression models and spread strategies; researchers studying seasonal gas market dynamics and LNG import trends.

What you get

Deliverable	Description
11 Jupyter notebooks	Storage EDA, seasonal decomposition, injection pace, winter adequacy, TTF curve, calendar spreads, price volatility, GARCH/HMM regimes, LNG analysis
Automated PDF report	Notebook 07 exports an 8-section PDF (storage snapshot, TTF curve, W-S spread, regression, injection pace, winter adequacy, spread analysis, volatility, LNG)
Python modules in src/	Reusable functions: AGSIClient, ALSIClient, DatabentoTTFClient, rolling_volatility, garch_volatility, spread_analysis, adequacy model

2 Quick Start (5 minutes)

Prerequisites

Python 3.11+, pip, JupyterLab

Installation

```
# Clone
git clone https://github.com/maziez11-lgtm/eu-gas-storage-analysis
cd eu-gas-storage-analysis

# Install dependencies (~2 min)
pip install -r requirements.txt

# Add API keys (see Section 3)
cp .env.example .env
# Edit .env: add AGSI_API_KEY and DATABENTO_API_KEY

# Launch JupyterLab
jupyter lab
```

First Run

Step	Action	Expected output
1	Open 01_data_ingestion.ipynb → paste AGSI key → Kernel → Restart & Run All	■ Storage: 1826 rows fill: 72.4%
2	Open 07_ttf_storage_analysis.ipynb → paste Databento key in cell 2 → Restart & Run All	■ Report: data/processed/gas_storage_ttf_report.pdf
3	In NB 07 last cell click '■ Download'	PDF opens / downloads

Tip: API keys are pasted directly into the notebook config cell (AGSI_API_KEY = "paste_key_here"). There is no dotenv runtime dependency.

3 API Keys

Service	URL	Cost	What it provides
AGSI+	agsi.gie.eu	Free	EU gas storage: fill rate, injection, withdrawal (TWh / GWh/day)
ALSI+	agsi.gie.eu	Free (same key as AGSI)	EU LNG terminal storage: fill rate, send-out (TWh / GWh/day)
Databento	databento.com	\$125 free credits; full history ~\$0.01	TTF futures M1–M24 daily settlement prices (€/MWh)

AGSI+ / ALSI+ key

1. Register at agsi.gie.eu
2. Log in → profile → copy API key
3. Paste into `.env` as **AGSI_API_KEY=your_key_here**
4. The same key works for both AGSI (gas storage) and ALSI (LNG)

Databento key

1. Register at databento.com/signup (credit card required; not charged within free credits)
2. Portal → API Keys → copy key starting with **db-**
3. Paste into `.env` as **DATABENTO_API_KEY=db-xxx...**

```
# .env
AGSI_API_KEY=your_agsi_key_here
DATABENTO_API_KEY=db-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

Note: Notebooks 01–06 and 11 need only the AGSI key. Notebooks 07–10 additionally need the Databento key to fetch the TTF forward curve.

4 Notebooks Reference

Run order: NB 01 must run first. NB 07 before NB 06, 08, 09, 10. NB 11 is independent.

#	Notebook	Purpose	Prerequisites	Key outputs	Runtime
01	data_ingestion	Fetch & cache AGSI+ EU + country storage	AGSI key	eu_aggregate_full.parquet	2–5 min
02	eda_storage_levels	YoY fill, 5yr bands, country heatmap	NB 01	Interactive Plotly charts	30 s
03	seasonal_analysis	STL decomposition, injection season summary, YoY delta	NB 01	STL chart, season table	1 min
04	injection_pace_tracker	90% Nov 1 target — required rate, achievability	NB 01	Required GWh/day, trajectory chart	30 s
05	winter_adequacy	4 demand × 3 injection scenarios, depletion	NB 01	Depletion curves, days-of-supply table	30 s
06	ttf_correlation	Rolling 30/60/90d fill vs TTF M1 correlation	NB 01, NB 07	Correlation series, scatter, OLS	30 s
07	ttf_storage_analysis	Integrated: curve, W-S spread, regression, PDF export	NB 01, Databento key	PDF report, ttf_curve.csv	3–5 min
08	time_spread_analysis	Calendar spreads with real month labels	NB 07	Spread matrix, regime table, streaks	1 min
09	ttf_price_analysis	Rolling vol, GARCH(1,1), price by fill, HMM regimes	NB 07, NB 01	Vol chart, regime labels, dist. table	2 min
10	spread_deep_dive	Roll yield, seasonality, animated curve slider	NB 07, NB 01	Roll yield series, animated chart	1 min
11	lng_storage_analysis	EU LNG fill, send-out, gas+LNG buffer	NB 01, ALSI key	eu_lng_full.parquet, LNG charts	2–5 min

5 Key Concepts

Storage Fill Rate

$$\text{fill (\%)} = \text{gasInStorage (TWh)} / \text{workingGasVolume (TWh)} \times 100$$

The EU regulatory target is **90% fill by November 1** each year (EU Regulation 2022/1032). At 90% the EU holds ~950–1,000 TWh entering winter.

API column	Unit	Description
gasInStorage	TWh	Working gas currently in storage
workingGasVolume	TWh	Total usable capacity
injection	GWh/day	Gas injected (positive flow)
withdrawal	GWh/day	Gas withdrawn (positive flow)
full	% (0–100)	Fill rate — auto-computed if absent

Unit conversion: injection and withdrawal are GWh/day; gasInStorage is TWh. To convert TWh → GWh multiply by 1,000.

TTF Forward Curve

TTF (Title Transfer Facility) is the primary European gas benchmark. Columns M1–M12 represent monthly settlement prices (€/MWh), where M1 = front month and M12 = 12 months forward.

Term	Months	Season
M1	Front month (nearest delivery)	Rolls ~3 days before expiry
Summer	Q2 + Q3	April – September
Winter	Q4 + Q1 next year	October – March
W–S spread	Winter price – Summer price	Key injection signal

Contango vs Backwardation

Regime	M1 – M2	Signal	Storage economics
Contango	Negative	Ample near-term supply	Injection profitable (buy spot, sell forward)
Flat	~0 (±€0.15)	Balanced	Marginal injection economics
Backwardation	Positive	Near-term tight; winter priced	Injection uneconomic; withdraw to sell now

Injection breakeven: Winter–Summer spread must exceed ~€5/MWh to cover underground storage costs (operations, gas losses, financing).

Roll yield: $(M1 - M2) / M1 \times (252 / \text{holding_days}) \times 100$. Positive = backwardation premium. Negative = contango roll cost.

Winter Adequacy Model

Scenario	Net withdrawal	Cold spell	Use case
Mild	4,500 GWh/day	None	Unusually warm winter
Normal	6,000 GWh/day	None	Historical average demand
Cold	7,500 GWh/day	14 days at 1.8×	Cold winter with snap
Extreme	9,000 GWh/day	21 days at 2.0×	2021-style cold event

Daily supply offsets: ~400 GWh/day LNG + ~800 GWh/day pipeline (configurable). Model runs 151 days, Nov 1 → Mar 31.

Rolling Correlation

Range	Interpretation
$r < -0.7$	Storage is the dominant price driver — normal regime
-0.7 to -0.3	Moderate influence; LNG / geopolitics competing
$r > -0.3$	Storage has lost explanatory power (supply shock, disruption)

LNG Columns (ALSI+)

Column	Unit	Description
gasInStorage	TWh	LNG in terminal tanks
dtmi	TWh	Declared total maximum inventory (capacity)
sendOut	GWh/day	Regasified gas sent to the grid
full	%	Fill rate = $\text{gasInStorage} / \text{dtmi} \times 100$

6 Example Outputs

Notebook 01 — Data Ingestion

```

■ Root: /home/user/eu-gas-storage-analysis
Fetching EU aggregate...
  DE: 1826 rows | 2020-01-01 → 2025-11-14
  FR: 1826 rows | 2020-01-01 → 2025-11-14
  ...
■ Storage: 1826 rows | 2020-01-01 → 2025-11-14
  Analysis date   : 2025-11-14
  Fill rate      : 72.4%
  Storage        : 928.1 TWh
  Capacity       : 1082.6 TWh

```

Notebook 07 — Integrated Analysis

```

■ Root: /home/user/eu-gas-storage-analysis
■ Storage: 1826 rows | latest: 2025-11-14 | fill: 72.4%
■ TTF curve loaded: 1512 rows | 2020-01-01 → 2025-11-14
  M1: €38.42/MWh  M3: €41.18/MWh  M6: €44.91/MWh  M12: €46.23/MWh
  Winter-Summer spread: +€6.49/MWh (contango)

OLS: log(TTF M1) ~ α + β × fill%
  α = 5.8241  β = -0.0243  R² = 0.614
  A 1pp increase in fill = 2.43% decrease in TTF (R²=0.61)

Injection scenarios: Low=6,240 / Avg=8,810 / High=11,380 GWh/day
  Low  → Nov 1: 79.3% ■ -10.7pp vs target
  Avg  → Nov 1: 89.6% ✓  -0.4pp vs target
  High → Nov 1: 97.2% ✓   +7.2pp vs target

■ 9 charts saved
■ Report: data/processed/gas_storage_ttf_report.pdf [847 KB]

```

Notebook 08 — Time Spread Analysis

```

Analysis date: 2025-11-14
TTF rows: 1512 | Storage rows: 1826

Spread matrix: 66 column pairs built
Latest M1-M2 spread: -€2.77/MWh (contango)
Latest M1-M12 spread: -€7.81/MWh

Longest streaks:
  start      end      regime      days
  2021-09-14 2022-06-28 backwardation 287
  2022-07-01 2023-04-11 backwardation 284
  2020-01-01 2021-09-13 contango      621

```

Notebook 09 — TTF Price Volatility

Analysis date: 2025-11-14 | TTF rows: 1512, Storage rows: 1826

Latest vol_21d: 34.7% vol_63d: 28.3%
 GARCH(1,1) fitted on 1511 log-returns.
 Latest GARCH cond. vol: 31.2% (annualised)

regime_label
 Low 412
 Medium 680
 High 420
 dtype: int64

Price distribution by fill bucket:

bucket_label	count	mean	median	p10	p90
20-36%	87	89.4	84.1	44.2	148.3
36-52%	210	72.1	68.4	32.8	121.6
52-68%	531	42.7	38.9	24.1	74.2
68-84%	512	31.8	29.4	19.6	52.7
84-100%	172	27.3	25.8	17.4	43.1

Notebook 11 — LNG Storage

■ Root: /home/user/eu-gas-storage-analysis
 Setup complete. Run cell 1 to fetch LNG data.

■ Saved 1681 rows → data/processed/eu_lng_full.parquet

	gasInStorage	sendOut	dtmi	full
date				
2025-11-14	7.21	312.4	8.24	87.5

Analysis date: 2025-11-14
 Latest LNG fill: 87.5%
 1-year avg send-out: 287 GWh/day

Correlation (gas fill vs LNG fill): 0.341

7 Workflows

Workflow 1 — Weekly Market Brief (10 min)

Step	Action
1 — Fetch data	Open 01_data_ingestion.ipynb → Kernel → Restart & Run All → confirm latest date
2 — Generate report	Open 07_ttf_storage_analysis.ipynb → Restart & Run All → wait for PDF
3 — Download	Last cell: click '■ Download' link — or open data/processed/gas_storage_ttf_report.pdf

Output: 8-section PDF covering storage snapshot, TTF curve, W-S spread, regression, injection pace, winter adequacy, time spreads, volatility, LNG.

Workflow 2 — Winter Adequacy Tracking (5 min)

Step	Action	Alert trigger
1	Run NB 01 (data fetch)	
2	Open 04_injection_pace_tracker → Run All	Achievable: False → cannot reach 90%; check max-injection scenario
3	Open 05_winter_adequacy → Run All	storage_depleted: True in Normal scenario → high stress
4	Cross-check NB 07, Section 4 (injection chart)	Current path below 90% target line → accelerate injection

Trigger: run weekly Apr–Oct; daily if fill rate drops >2pp in a week.

Workflow 3 — Spread & Roll Yield Analysis (10 min)

Step	Action	Key metric
1	Run NB 07 (required — fetches TTF curve)	
2	NB 08 → Run All — read Section 1 W-S spread, Section 3 regime	W-S > €8/MWh → significant winter risk priced
3	NB 10 → Run All — Section 2 roll yield, Section 4 seasonality	Positive roll yield → backwardation premium
4	NB 10, Section 5 — animated curve slider: scrub to historical stress dates	Compare curve shape vs Sep 2021, Aug 2022

Workflow 4 — Historical Curve Comparison (5 min)

```
# In NB 07 Section 6 (Interactive Curve Tool):
CURVE_DATE = date(2022, 8, 26) # TTF peak above €300/MWh
# Run the cell → see historical curve vs today's shape

# Or in NB 10 Section 5 — animated slider:
# Click ■ Play or drag to comparison date
```

Workflow 5 — Incremental Data Update (3 min)

```
from src.agsi_client.databento_client import DatabentoTTFCClient

# TTF curve – fetch only new rows since last CSV date
ttf_client = DatabentoTTFCClient(api_key='your_key')
df = ttf_client.update_ttf_csv('data/raw/ttf_curve.csv', n_months=12)
# Logs: 'Last date: 2025-11-07 | Fetching from 2025-11-08'
# Logs: '+5 new rows'
# Logs: '■ Saved 1517 rows → data/raw/ttf_curve.csv'

# Storage – just re-run NB 01 (cache handles deduplication, 12h TTL)
```

Schedule: run `update_ttf_csv` daily after ~18:00 CET (ICE settlement time).

8 Data Files

Filename	Location	Created by	Contents	~Rows
eu_aggregate_full.parquet	data/processed/	NB 01	EU daily: gasInStorage, injection, withdrawal, workingGasVolume, full	1,500–2,000
eu_lng_full.parquet	data/processed/	NB 11	EU LNG daily: gasInStorage, sendOut, dtmi, full	1,500–2,000
tff_curve.csv	data/raw/	NB 07	TTF forward curve: columns M1–M12, daily settlement (€/MWh)	1,500–2,000
gas_storage_ttf_report.pdf	data/processed/	NB 07	8-section PDF report	—
*.parquet (cache)	data/cache/	Auto (AGSIClient)	Per-request AGSI API response cache, 12h TTL	—
*.parquet (cache)	data/cache/alsi/	Auto (ALSIClient)	Per-request ALSI API response cache, 12h TTL	—

Loading parquet files

```
import pandas as pd

df = pd.read_parquet('data/processed/eu_aggregate_full.parquet')
df.index = pd.to_datetime(df.index).tz_localize(None) # strip tz
df = df.sort_index()
```

Cache management

```
# Force fresh fetch (bypass cache)
df = client.get_eu_aggregate(start='2020-01-01', use_cache=False)

# Clear all cache files
client.clear_cache()
```

9 Troubleshooting

Error / Symptom	Cause	Fix
<code>TypeError: Cannot join tz-naive and tz-aware DatetimeIndex</code>	API returns UTC-aware timestamps; storage parquet may be tz-naive	Add <code>.tz_localize(None): df.index = pd.to_datetime(df.index).tz_localize(None)</code>
<code>ModuleNotFoundError: No module named 'src'</code>	Notebook opened from wrong directory; project root not in <code>sys.path</code>	Run the path-fix cell at the top of every notebook first (the <code>for _c in [...]</code> block)
<code>reversed()</code> argument must be a sequence (Python 3.14+)	Python 3.14 tightened <code>reversed()</code> protocol; some Plotly internals affected	Replace <code>reversed(x)</code> with <code>x[::-1]</code> in any custom code; <code>src/visualization/plots.py</code> already uses slice notation
ALSI sendOut / dtmi columns missing or all NaN	Not all LNG terminals report all fields to GIE	<code>ALSIClient._parse()</code> coerces to NaN; guard with <code>lat.get('sendOut', float('nan'))</code>
Only 300 rows returned from AGSI / ALSI	API paginates at 300 rows per page	Clients handle pagination automatically. If truncated, verify API key — invalid keys often return one empty page
<code>FileNotFoundError: eu_aggregate_full.parquet</code>	Notebook 01 has not been run yet	Run <code>01_data_ingestion.ipynb</code> (Kernel → Restart & Run All) before any other notebook
<code>FileNotFoundError: ttf_curve.csv</code>	Notebook 07 cell 2 (Databento fetch) has not been run	Open NB 07, run cell 2 with a valid <code>DATABENTO_API_KEY</code>
<code>ImportError: No module named 'arch'</code>	Optional GARCH dependency not installed	<code>pip install arch</code> (already in <code>requirements.txt</code> — re-run <code>pip install -r requirements.txt</code>)
Charts blank / kaleido error in PDF export	kaleido not installed or incompatible version	<code>pip install kaleido==0.2.1</code> (must match Plotly version)
Cache returning stale data	12h TTL not yet expired	Pass <code>use_cache=False</code> to any client method, or call <code>client.clear_cache()</code>

EU Gas Storage Analysis · github.com/maziez11-lgtm/eu-gas-storage-analysis · Data: GIE AGSI+ · GIE ALSI+ · Databento NDEX.IMPACT